Molecular insight into the nanoconfined calcite–solution interface

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Little is known about the influence of nanoconfinement on calcium carbonate mineralization. Here, colloidal probe atomic force microscopy is used to confine the calcite–solution interface with a silica microsphere and to measure Derjaguin–Landau–Verwey–Overbeek (DLVO) and non-DLVO forces as a function of the calcium concentration, also after charge reversal of both surfaces occurs. Through the statistical analysis of the oscillatory component of a strong hydration force, the subnanometer interfacial structure of the confined atomically flat calcite is resolved in aqueous solution. By applying a mechanical work, both water and hydrated counterions are squeezed out from the nanoconfined solution, leaving the calcite surface more negatively charged than the analogous unconfined surfaces. Layer size and applied work allow a distinction between the hydration states of the counterions in the Stern layer; we propose counterions to be inner- and outer-sphere calcium ions, with a population of inner-sphere calcium ions larger than on unconfined calcite surfaces. It is also shown that the composition of the nanoconfined solution can be tuned by varying calcium concentration. This is a fundamental study of DLVO and hydration forces, and of their connection, on atomically flat calcite. More broadly, our work scrutinizes the greatly unexplored relation between surface science and confined mineralization, with implications for diverse areas of inquiry, such as nanoconfined biomineralization, CO_2 sequestration in porous aquifers, and pressure solution and crystallization in confined hydrosystems.

calcite | hydration forces | DLVO theory | nanoconfinement | atomic force microscopy

Understanding the effect of nanoconfinement on the solution composition near the calcite surface is crucial in revealing the mechanisms of biomineralization in confinement, because the confined thin film of aqueous electrolyte, which provides the path for ions and water to the buried mineral interface, can behave totally different from the unconfined (free) solution.

Whereas the effect of confinement on the properties of the calcite–solution interface is largely unexplored, the Stern layer of unconfined calcite surfaces has been intensively investigated via simulations and experiments. Interestingly, Ca^{2+} and CO_3^{2-} (and HCO_3^-) ions are not adsorbed directly to calcite but to surface-adsorbed water molecules as outer-sphere species (OS) due to calcite's strong affinity to water (1). X-ray reflectivity studies (2, 3) and molecular dynamics (MD) simulations (4) have proved that the calcite–solution interface is well defined by two layers of water molecules. Hence, instead of the conventional model, where the Stern layer is adjacent to the charged surface, calcite's Stern layer, the compact layer of counterions, is believed to be located on top of two water layers: OS calcium (OS-Ca^{2+}) ions mainly populate the interface at ∼4.9 Å from the unconfined calcite surface, whereas the population of inner-sphere calcium (IS-Ca^{2+}) ions is located at ∼3.3 Å from the calcite surface and is less energetically favorable than the population of OS-Ca^{2+} ions (5).

The focus of this work is to investigate the nanoconfined calcite–solution interface via force measurements. The Derjaguin–Landau–Verwey–Overbeek (DLVO) theory (6, 7), which includes van der Waals (vdW) and electrical double-layer (EDL) forces (8), is commonly used as a benchmark to discuss the forces between surfaces that are slightly charged in dilute electrolyte solutions. However, the DLVO theory does not predict ion-specific effects, and, being a continuum theory, it must fail at the molecular scale. Deviations are known to occur at small surface separations (<5 nm), where the role of ion–ion correlations, ion-specific effects, ion hydration, and solvent structure are no longer negligible (8, 9). Discrepancies from the DLVO theory have been reported at concentrations as low as 0.3 mM (10): Direct surface force measurements have revealed repulsive hydration forces with decay lengths ranging from 0.1 to 1 nm (10, 11) with an oscillatory component, first considered to result from the oscillatory density profile (i.e., layering) of confined water layers (12). Besides water layering, the surface adsorption of hydrated ions (13, 14) and, more recently, their confinement-induced layering (15, 16), have been also considered to be responsible for the repulsive hydration forces. The experimental results by Pashley first demonstrated the ion specificity of the hydration force (10).

Only recently the surface forces between two calcite surfaces were investigated by atomic force microscopy (AFM) (17). Despite the beneficial chemical symmetry of this system, the uncontrolled geometry of the probe, a calcite piece, led to a contact topography with an undefined effective radius, which hindered modeling and scrutinization of surface interactions. Further, unpredictable asperities in the contact area may lower the experimental resolution, which could explain the absence of oscillatory hydration forces in these measurements and thereby, the lack of molecular insight into the confined solution.

In this work, colloidal probe AFM was used to measure the surface forces between calcite and a silica colloid, ensuring a well-defined and reproducible contact geometry (18). Despite the complexity added by dissimilar surfaces, this technique has been

Significance

Calcium carbonate mineralization occurring in confined spaces and nanopores is central to many natural processes occurring at or near the Earth's surface. We study the effect of divalent ion concentration, in an attempt to provide a general conceptual picture of Derjaguin–Landau–Verwey–Overbeek (DLVO) and non-DLVO forces considering calcite's nonclassical Stern layer. From the careful measurement of the hydration forces in aqueous solutions, we propose a model for the nanoconfined calcite–solution interface. Colloidal probe AFM resolves subnanometer film thickness transitions of the nanoconfined aqueous solution induced by the squeezing out of water and hydrated calcium ions by applying an external work. The fundamental knowledge derived from this work provides guidance to understand the physicochemical phenomena occurring at nanoconfined mineral interfaces.

Author contributions: R.M.E.-M. designed research; Y.D. and R.M.E.-M. performed research; Y.D. and R.M.E.-M. analyzed data; R.M.E.-M. supervised the graduate student; and Y.D. and R.M.E.-M. wrote the paper.

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proven to be reliable and powerful to directly measure surface forces (19, 20). The DLVO theory describes well the long-range interactions between the confining surfaces despite calcite's unconventional Stern layer. We study the subnanometer interfacial structure of calcite in solution at the molecular level through the statistical analysis of the hydration forces and compare it to that of unconfined calcite, in an attempt to provide a starting point for a new conceptual understanding of the nanoconfined calcite–solution interface and of nanoconfined mineralization.

Results

Force–Separation Curves for Calcite–Silica Systems. Calcite crystals ($10\bar{1}4$) were equilibrated for 24 h in $CaCl_2$ solution at the selected concentrations of 0, 1, 10, and 100 mM. After equilibration, the solution was saturated with respect to calcite (see concentration and pH in *SI Appendix*, Table S1). Silica microspheres glued to AFM tipless cantilevers were immersed in the $CaCl_2$ solutions previously saturated with calcite (termed "$CaCl_2$/$CaCO_3$ solutions") for at least ∼3 h before force measurements started, which were prolonged over 8 h. All silica spheres were imaged (*SI Appendix*, Fig. S1) and only used if the rms roughness was smaller than 2 nm.

Force measurements were performed by moving the cantilever toward an atomically flat region of the calcite crystal on the ($10\bar{1}4$) plane at a velocity of 20 nm/s to avoid hydrodynamic effects (see negligible drag in *SI Appendix*, Fig. S2). Due to the inherent uncertainty of the absolute tip–substrate separation in AFM force measurements, we note that the abscissa has an arbitrary zero but we refer to it as "separation" (D). To relate the measured forces (F) to the thermodynamic energy change per unit area between flat surfaces according to the Derjaguin approximation (21), the force is normalized by the effective radius as F/R_{eff}, with $R_{eff} = R$ for the plane-sphere geometry. The use of F/R_{eff}, instead of F, is widely practiced in surface force studies to allow the comparison across different systems. The normalized forces are presented as force–separation curves.

vdW and EDL forces were modeled according to the DLVO theory at separations larger than 5 nm. Modeling details are described in *SI Appendix*. The Lifshitz theory was used to estimate the Hamaker constant ($A = 6.95*10^{-21}$ J). Both calcite and silica have low surface potentials in the selected solutions according to zeta potential measurements (*SI Appendix*, Fig. S3). At low surface potentials (<14 mV), an analytical expression for the EDL force, which is valid for dissimilar confining surfaces and mixtures of monovalent and multivalent ions, is obtained by solving the linearized Poisson–Boltzmann equation (8, 19). The fits are satisfactory assuming constant regulation, which considers that neither the diffuse layer potential nor the surface charge remains constant with surface separation; the constant regulation parameter gives the relation between diffuse and Stern layer capacitance and is dictated by the dissociation reactions occurring at the surface. We did not attempt to manually shift the (estimated) outer-Helmholtz plane to fit the EDL force because the true surface separation is unknown anyway in AFM force measurements. Hence, the obtained parameters are a rough estimation and they need to be considered with caution.

Fig. 1*A* shows representative force–separation curves as a function of $CaCl_2$ concentration in $CaCl_2$/$CaCO_3$ solutions; the solid lines show the fit of the DLVO theory to the measured force–separation profiles at the concentrations of 0 and 1 mM. In 0 mM, i.e., in saturated calcium carbonate solution, force–separation curves exhibit a strong short-range repulsion, consistent with reported hydration forces (10, 15), and a longer-range repulsive force that decays with the exponential function of the separation (D) as expected for the EDL force. The addition of $CaCl_2$ (1 mM) leads to an increase of the hydration repulsion, while the EDL force range slightly decreases, which is attributed to the decrease of the Debye length (DL) from 7.70 to 4.79 nm.

A closer look at two different force profiles at 10 mM (DL = 1.76 nm) reveals a twofold behavior at this concentration. We observed a weak attractive force (red in Fig. 1*B*), but occasionally, a weak repulsion (*SI Appendix*, Fig. S4). The DLVO theory gives a positive potential for silica (∼+58 mV; *SI Appendix*, Table S3), whereas the surface potential of calcite is close to neutral but slightly negative (−1.8 ± 1.4 mV). The obtained surface potentials can qualitatively explain a weak EDL attraction for a slightly negative potential and a weak EDL repulsion for neutral calcite.

By further increasing the concentration from 10 to 100 mM, the EDL totally collapses (DL = 0.58 nm); the secondary minimum (8) is very weak and often not detected, which is partially attributed to the superposed strong hydration repulsion. Recently (22), an increasing DL with solution concentration was shown for electrolyte solutions at high concentration ($d/DL>1$, with d, the mean diameter of the unhydrated ion); the origin of this repulsion is not understood yet. Nevertheless, this high-concentration regime is not relevant for the present study, because d/DL is ∼0.56 at 100 mM.

Fig. 1*B* shows steps in the force–separation curves for the closest 3 nm of separation between calcite and silica colloid. These steps are reminiscent of the oscillatory component of the hydration force (10–12) or of any solvation force (8) between two atomically flat

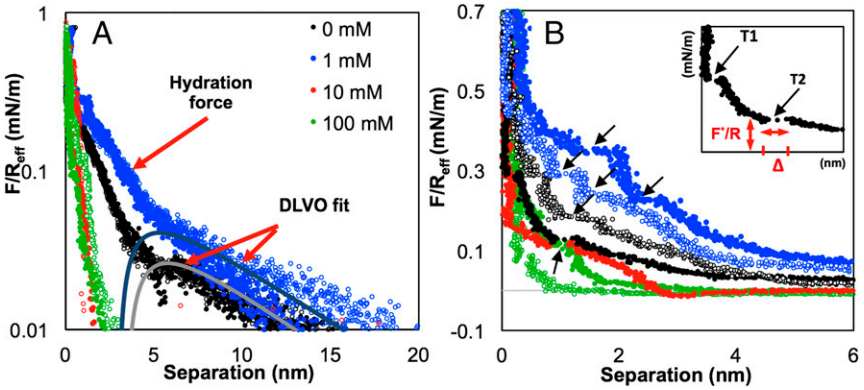


Fig. 1. Representative force–separation curves between calcite and a silica colloid in $CaCl_2$/$CaCO_3$ solutions at separations (*A*) smaller than 20 nm and (*B*) smaller than 6 nm. The color scheme is consistent throughout this work: black for 0 mM, blue for 1 mM, red for 10 mM, and green for 100 mM $CaCl_2$. Fits based on the DLVO theory lead to surface potentials for calcite, $\psi_c = -12.8 \pm 2.2$ mV, -12.2 ± 2.5 mV, and -1.8 ± 1.4 mV, and for silica, $\psi_s = -6.3 \pm 0.9$ mV, -7.6 ± 2.3 mV, and $+57.8 \pm 12.4$ mV, at 0, 1, and 10 mM, respectively, and regulation parameters $p_c = 0.62 \pm 0.07$ and $p_s = 0.88 \pm 0.07$. The steps in the repulsive force (indicated by arrows in *B*) result from the squeezing of layers of water and hydrated ions. (*B*, *Inset*) shows the FTT closer to the hard wall (T1), the subsequent FTT (T2) at a larger surface separation, the layer thickness (Δ), and the layering force (F^*/R_{eff}).

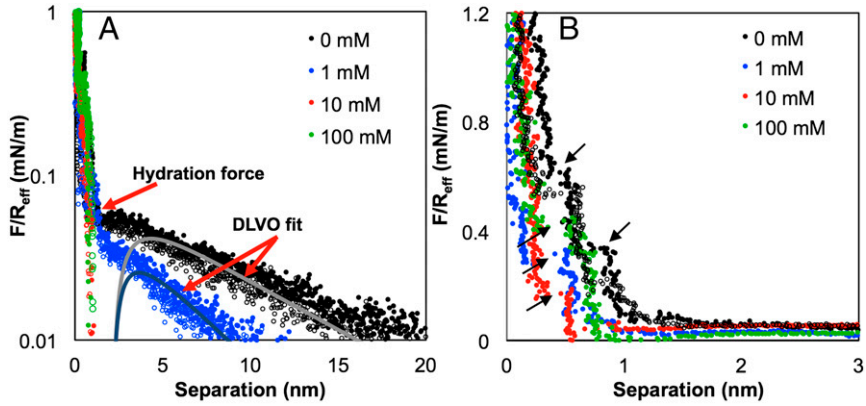


Fig. 2. Representative force–separation curves between two silica colloids (R_{eff} = R/2 = 2.5 μm for the sphere–sphere geometry) in $CaCl_2/CaCO_3$ solutions at separations (*A*) smaller than 20 nm and (*B*) smaller than 3 nm. The arrows in *B* point at FTTs. The fits according to the DLVO theory lead to a surface potential equal to −8.3 ± 0.4 mV for saturated $CaCO_3$ solution (0 mM $CaCl_2$), −6.7 ± 0.7 mV for 1 mM $CaCl_2$, and +14.2 ± 1.0 mV for 10 mM $CaCl_2$, and a regulation parameter p_s = 0.88 ± 0.07. The x-axis range is selected to be smaller than in Fig. 1*B* to help to identify the steps in the hydration force.

surfaces. As a large number of molecules throughout the confined film are collectively squeezed out from the confined region (as a layer), the surface jumps to the next stable position. We denote these discontinuities in the repulsive force as "film-thickness transitions" (FTTs) and the load applied to force the transition as "layering force." Owing to the smaller lateral confinement, the FTTs are less pronounced than in our previous surface forces apparatus measurements on mica (15), but are clearly detectable. Hence, these results demonstrate that AFM resolves multiple FTTs (i.e., layers) upon confinement of an atomically flat calcite surface. A detailed scrutiny of the thickness of the layers (Δ) and the layering force (F^*/R_{eff}) is given in *Statistical Analysis*.

Force–Separation Curves for Silica–Silica Systems. AFM force measurements were performed on silica–silica (symmetric) systems for comparison. Silica colloids glued to silicon wafers and to AFM cantilevers were submerged in the investigated $CaCl_2/CaCO_3$ solutions for at least 3 h before force measurements. The reactivity of the silica surface has been shown to influence force measurements in previous studies through the gradual formation of a gel on the silica surface with time (23, 24). This phenomenon was occasionally observed in silica–silica systems at a concentration of 0 mM (*SI Appendix*, Fig. S5), but these results have been excluded from this discussion.

Fig. 2*A* shows representative force–separation curves. The surface forces were modeled by the DLVO theory with an experimental Hamaker constant of 2.8×10^{-21} J (20) for separations larger than 5 nm and under the assumption of constant charge regulation with an effective radius of R_{eff} = R/2 (see details in *SI Appendix*). The EDL significantly collapses in both 10 and 100 mM, in agreement with previous work with calcium and other multivalent ions (20, 25). Hydration forces with at most two FTTs were resolved at all of the investigated concentrations (Fig. 2*B*). It is noteworthy that the hydration force between silica colloids is weaker and of shorter range than between calcite and silica at all concentrations; note the different scales in Figs. 1*B* and 2*B*.

Statistical Analysis. Fig. 3 shows 2D histograms with the collection of normalized layering force (F^*/R_{eff}), i.e., the force required for the FTT to occur, and the thickness of the FTT ($\Delta = D_2 - D_1$). The observed size of an FTT (Δ) represents the difference between the film thickness before (D_2) and after (D_1) the transition, but it is generally assigned to the size of a particular molecule or complex that arranges in a layer at the confined interface. The magnitude of the layering force (F^*/R_{eff}) is a measure of the strength of the interaction; a higher force indicates a stronger adsorption of the molecules either to the surface or to the underlying molecules. Hence, the FTTs in the hydration force reflect the molecular structure of the nanoconfined film of solution. Fig. 3 *A* and *C* shows the FTT closer to the hard wall (denoted as T1), and Fig. 3 *B* and *D* shows the rest of the FTTs (denoted as T2+). The distribution of layering forces and layer thicknesses was fitted to multipeak Gaussian distributions by Igor Pro to determine the peak means, widths, and relative frequencies (*SI Appendix*, Tables S4 and S5). We limited the maximum load to 4 nN (Hertzian pressure of 24 MPa) to

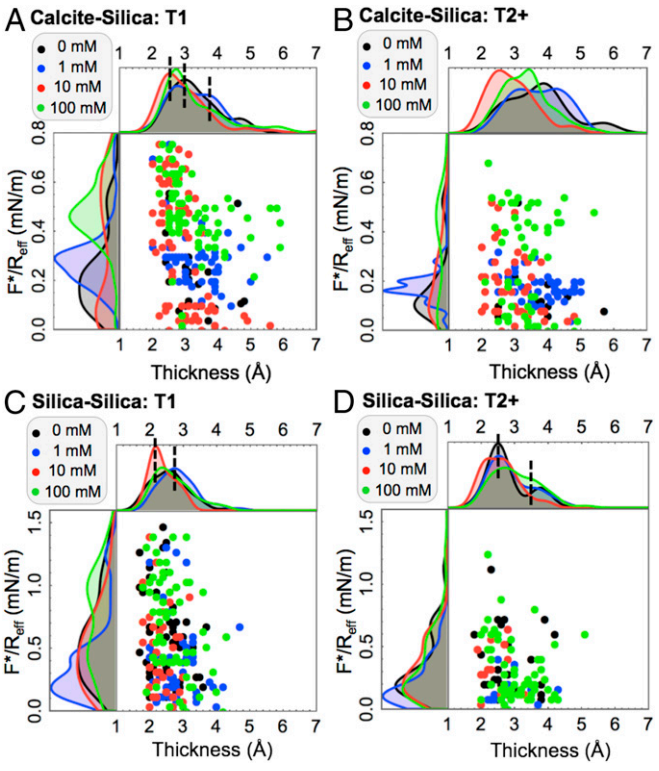


Fig. 3. Two-dimensional histograms of the normalized layering force (F^*/R_{eff}) vs. layer thickness (Δ) resolved in the hydration force at the selected concentrations of $CaCl_2/CaCO_3$ solutions on calcite–silica systems (*A*) for the first FTT (T1, closer to the hard wall) and (*B*) for all the other FTTs (T2+), and on silica–silica systems (*C*) for the first FTT (T1) and (*D*) for all other FTTs (T2+). The black dashed lines point at frequent positions.

minimize the effect of pressure dissolution of calcite. Admittedly, the definition of T1 remains arbitrary because it refers to the last experimentally resolved layer before reversing the cantilever direction; one can never exclude the occurrence of additional transitions at higher loads.

The 2D histogram in Fig. 3*A* shows the distribution of layer thicknesses for T1 in calcite–silica systems. At 0 mM, i.e., in saturated calcium carbonate solution, the layer thickness has a major peak at $\Delta \sim 3.0 \pm 0.6$ Å, and less-pronounced peaks at $\Delta \sim 2.6$ Å and $\Delta \sim 4.8$ Å. By adding 1 mM $CaCl_2$, two characteristic peaks are detected for T1 with similar frequency at 2.7 ± 0.5 Å and 3.7 ± 0.6 Å. By further increasing the concentration to 10 mM, a trimodal distribution is measured ($\Delta \sim 2.6 \pm 0.4$ Å, 3.3 ± 0.5 Å, and 5.0 ± 0.5 Å), but the frequency of the 2.6-Å transition and the associated applied force are clearly higher. Similarly, at 100 mM $CaCl_2$, three distinct layer thicknesses are resolved: 2.7 ± 0.3 Å, with larger layering force, and 3.7 ± 0.4 Å and 5.4 ± 0.8 Å, at lower layering forces. Note that these measurements cannot distinguish between the size of hydrated anions and cations. Nevertheless, considering the negative surface charge, it is expected that more counterions (calcium) populate the interfacial region. If charge reversal takes place, a higher near-surface concentration of anions and higher ion correlations can be expected (26).

Fig. 3*B* shows the 2D histograms for T2+ in calcite–silica systems, which takes place at a larger surface separation than T1, and hence, at a lower force (Fig. 1*B*, *Inset*). In saturated calcium carbonate solution (0 mM), Δ is $\sim 4.0 \pm 0.6$ Å and $\sim 2.7 \pm 0.6$ Å. By increasing the concentration to 1 mM, two characteristic sizes are equally likely ($\Delta \sim 3.0 \pm 0.6$ Å and 4.2 ± 0.7 Å), the latter with a slightly lower layering force. At 100 mM, the trimodal distribution is recovered ($\Delta \sim 2.7 \pm 0.6$ Å, 3.5 ± 0.2 Å, and 4.2 ± 0.3 Å), with the highest frequency for $\Delta \sim 3.5$ Å. In contrast, at 10 mM the peaks at 2.4 ± 0.4 Å and 3.2 ± 0.7 Å are most pronounced, whereas FTTs with thickness larger than 4 Å are much less pronounced than at other concentrations.

A similar analysis was performed on silica–silica systems for comparison. Fig. 3*C* shows that the layer thickness distribution of T1 is narrower than for calcite–silica systems, collectively shifting to smaller sizes. In saturated calcium carbonate solution (0 mM), the resolved layer has a characteristic thickness of ~2.7 Å, similar to that in 1 mM $CaCl_2$. By increasing the concentration to 10 mM, a smaller characteristic thickness of ~2.2 Å becomes more preeminent than that of ~2.7 Å. Similarly, at 100 mM, the most characteristic layer size is $\Delta \sim 2.3$ Å, whereas $\Delta \sim 3.0$ Å is less frequent. Fig. 3*D* shows a bimodal distribution for the layer thickness of T2+ with characteristic peaks at ~2.5 Å and ~3.5 Å, the latter with a much smaller frequency. Although a small peak at >4 Å is observed at all concentrations, the frequency is notably smaller than in calcite–silica systems.

Discussion

This study extends the general picture of DLVO and non-DLVO forces to surfaces with a nonconventional Stern-layer, here to calcite. Fig. 1*A* shows that the EDL force can be predicted by the DLVO theory at large separations, taking into account that the surface potential can deviate from that of the unconfined surfaces. Under strong confinement, however, the hydration force reveals the composition of the strongly collapsed EDLs at both confining surfaces. Thus, the near-surface concentration of water and (hydrated) ions not only determines the Stern layer, and thereby the EDL force, but also the onset of the hydration force, its magnitude, and FTTs. We generally observe both a significant increase in the occurrence of multiple FTTs (i.e., layers) and a shift to higher layering forces with increasing concentration. This is consistent with an increasing near-surface concentration of Ca^{2+} ions, which implies a higher energy to squeeze out the electrostatically attracted counterions to the carbonate oxygen at the surface. Further, the increase in surface concentration of Ca^{2+} ions with bulk concentration obtained through the analysis of the FTTs is consistent with the less-negative surface potentials of silica and calcite obtained by fitting the DLVO forces (*SI Appendix*, Tables S2 and S3).

Although the concentration is expected to influence both the water structure (27) and the size of ion hydration shells (28), we do not attempt to assign a different composition to the multiple peaks shown in Fig. 3 due to the overlapped associated errors. Based on the statistical analysis, we identify three main characteristic layer thicknesses on calcite–silica systems: $\Delta \sim 2.4$–2.8 Å, $\Delta \sim 3$–4 Å, and $\Delta \sim 4$–5 Å. The richness in Δ-values is attributed to the presence of water, and IS-Ca^{2+} and OS-Ca^{2+} ions at the confined calcite interface, which are characterized by different size (Δ), adsorption energies (5), and applied work, as discussed below. Unlike hard-sphere model fluids, such as octamethylcyclotetrasiloxane, which create periodic oscillatory forces between hard surfaces (8, 29), the layering in aqueous electrolyte solutions is more like a layering of "soft spheres," because the distinction between hydration water and free water is energetically not sharp, mainly due to hydrogen bonding. This may contribute to the broadening of the distributions.

The diameter of the primary hydration shell of OS-Ca^{2+} ions is reported to be ~4.8–5.2 Å [with 8–10 water molecules (5)], and hence it is close to the size of the largest FTTs. Because the size of a layering transition is an estimate for the size of the layering objects, we can roughly estimate the varying effective coordination numbers for hydrated calcium with diameters in the range ~3–5 Å, considering the excluded volume of a spherical hydrated ion (15). This exercise results in coordination numbers as small as ~5 for $\Delta \sim 3.5$ Å, and hence closer to the reported values for IS-Ca^{2+} ions (~6.6) on unconfined calcite (5). The confinement-induced (partial) dehydration could be overestimated, but it is nevertheless in qualitative agreement with the MD simulations for nanoconfined NaCl solution (30) and the neutron diffraction study for $CaCl_2$ solution between vermiculite plates (31).

Fig. 4*A* shows a cartoon of the confined solution between calcite and silica based on the present study. Similar to the unconfined calcite–solution interface, the confined solution is composed of layers of water and hydrated ions; in force measurements they are squeezed out upon an applied load, thereby leading to steps (i.e., FTTs) in the hydration force. FTTs with a larger size ($\Delta \sim 3$–5 Å) are detected at larger surface separations and are attributed to layers of hydrated calcium ions (IS-Ca^{2+} for $\Delta \sim 3$–4 Å and OS-Ca^{2+} for $\Delta \sim 4$–5 Å), followed by a smaller transition ($\Delta \sim 2.4$–2.8 Å) that is closer to the size of water molecules. The collective shift to larger thicknesses from T1 to T2+ indicates that larger species, i.e., more hydrated calcium ions relative to either less-hydrated calcium ions or water molecules, populate the layers further away from the surfaces. In the example in Fig. 4*A*, layers

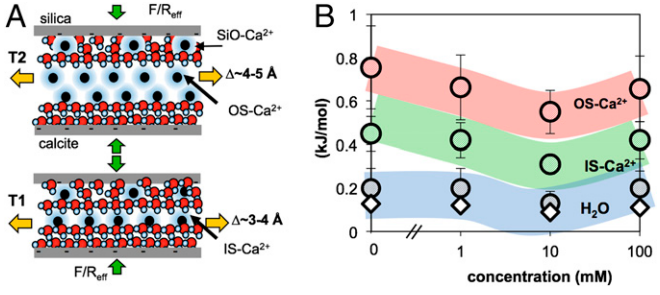


Fig. 4. Energetics of squeezing out a solution confined between calcite and silica upon applied load. (*A*) A cartoon for calcite–silica systems with a possible scenario. (*B*) The estimated work per mole applied to squeeze out OS-Ca^{2+}, IS-Ca^{2+} ions, and water molecules confined in calcite–silica systems (circles) and silica–silica systems (diamonds). The mean values were obtained from the 2D histogram constructed for the work (per mole) vs. layer thickness (*SI Appendix*, Fig. S6).

of OS-Ca^{2+} (*Top*) and IS-Ca^{2+} (*Bottom*) ions are resolved. Importantly, the population of IS-Ca^{2+} ions appears to be more significant in confinement than on unconfined calcite, perhaps due to the forced dehydration by the applied load.

At 10 mM $CaCl_2$, the population of IS-Ca^{2+} seems to dominate. A previous amplitude modulation AFM study in which the calcium ions were imaged on calcite surfaces also showed a different behavior at 10 mM, with calcium ions being easily pushed away by AFM sharp tips due to much more loose binding to calcite (32). A small EDL attraction or repulsion was measured at this concentration (Fig. 1*B*), and attributed to the interfacial structure of a near-neutral calcite surface. It is possible that the layer with $\Delta \sim 3.2$ Å composed of IS-Ca^{2+} ions provides the charge neutralization. This would lead to a weak adsorption of OS-Ca^{2+} ions, thereby being easily pushed away with the colloid and not always detectable, which would explain the increase in the relative frequency of layers of IS-Ca^{2+} ions at this concentration. Note that the small peak at $\Delta > 4.0$ Å (Fig. 3*B*), which is measured at a larger separation than the 3.2-Å transition, could correspond to the loosely bound OS-Ca^{2+} ions.

We can estimate the energetics involved in squeezing out the confined solution to further support our hypothesis. The measured layering force (F*) can be converted into the work applied to push away a layer of size Δ, as $W = F^* \cdot \Delta$. The confined region has a Hertzian radius of $a \sim (3F^*R_{eff}/4E^*)^{1/3}$ and the work per unit area is $W/\pi a^2$, which can be converted into applied work per mole, $E = WA_{mol}/\pi a^2$, where A_{mol} is the area occupied by 1 mole of molecules in a layer; A_{mol} is estimated as $\sim N_A \pi\,(\Delta/2)^2$, and N_A is the Avogadro number. We have constructed 2D histograms for the calculated work E vs. Δ (*SI Appendix*, Fig. S6). Fig. 4*B* summarizes the calculated mean values as a function of the concentration.

The three transitions (2.4–2.8 Å, 3–4 Å, and 4–5 Å) are energetically different. Higher work (per mole, not per area) is applied to squeeze out larger layers, and hence attributed to the squeezing out of hydrated counterions. This is consistent with the reported higher free energy of adsorption of OS-Ca^{2+} and IS-Ca^{2+} to calcite compared with that of water (5). Further, the applied work per mole follows the trend of the surface potential of confined calcite achieving a minimum at 10 mM, where calcite is close to neutral; we note that this agreement is surprisingly obtained by two completely independent analyses (DLVO theory and hydration force), which strengthens the connection between them. The highest energy is required in saturated calcium carbonate solution (0 mM), which might be attributed to the highest negative surface potential of calcite (and silica), and hence to the strongest electrostatic interactions between Ca^{2+} ions and calcite. Although the error bars are large, the applied work per mole tends to increase upon charge reversal. Such an increase could be originated by stronger ion–ion correlations in the confined solution at the higher surface concentrations.

Atomistic simulations on single calcite surfaces have shown that OS-Ca^{2+} ions need to overcome an energy barrier of $E_B \sim 14$ kJ/mol to adsorb onto $(10\bar{1}4)$ calcite surfaces to become IS-Ca^{2+} ions (5). This barrier is attributed to the electrostatic repulsion between the cation and the surface-water layers with an excess of hydrogen atoms. The applied work during force measurements, albeit roughly estimated, is one order of magnitude smaller than E_B. Accordingly, OS-Ca^{2+} ions are more likely to be pushed away than to be pushed closer to the surface as IS-Ca^{2+} ions. However, the high frequency of transitions with $\Delta \sim 3$–4 Å supports that the latter could happen, especially with increasing concentration.

The selected counter surface material, silica, influences the interfacial structure of the confined solution as well. The thickness of the FTTs in silica–silica systems is characteristically smaller than 3 Å. It is reasonable to expect that Ca^{2+} ions bridge surface deprotonated silanol groups (33), and thus they are not easily squeezed out. In fact, after thoroughly rinsing the silica colloids with water, Raman microspectroscopy still detects bound calcium to the silanol groups (*SI Appendix*, Fig. S7). Therefore, the resolved film thickness transitions are mainly attributed to the dehydration of the silica surfaces; at most two water layers are detected. Reported hydration energy of silanol is ~27 kJ/mol (33), whereas the applied work (~0.2 kJ/mol, diamonds in Fig. 4*B*) is ~2 orders of magnitude smaller. Interestingly, it is similar to the applied work in silica–calcite systems to squeeze out layers with a size smaller than 3 Å (blue circles), thereby suggesting that calcite could remain hydrated at the maximum applied load, whereas the resolved water layers would result from the dehydration of the silica surface. This is plausible because the reported values for the hydration energy of $(10\bar{1}4)$ calcite surfaces [~94 kJ/mol (34)] are even higher than those of silanol. Nevertheless, the surface dehydration induced by external work in confinement seems to be more energetically favorable than in free solution, perhaps because of the accompanying increase in entropy of the system involved in the release of the confined water molecules.

The cartoon in Fig. 4*A* is also helpful to describe mineral growth in confinement because the first step in growth is the adsorption of ion pairs and complexes to the mineral surface. The strongly repulsive hydration force provides a path for the ions to diffuse to the buried interface. The proposed composition of the confined solution implies the presence of a high concentration of calcium ions nearby the confined calcite–solution interface (e.g., $1/\Delta^3 \sim 1/5^3$ $Å^{-3} \sim 13$ M OS-Ca^{2+}). Although coions (Cl^-, CO_3^{2-}, and HCO_3^-) are neglected in the cartoon, it is expected that they are present within the confined solution as well, especially at high concentrations, where ion–ion correlations become more significant (32). Let us consider a hypothetical scenario in which the calcium carbonate concentration in the confined solution becomes high enough (supersaturated) for mineral growth. For an ion pair or carbonate–calcium complex to attach to the calcite surface, a dehydration barrier (for calcium carbonate and calcite surface) needs to be overcome, and the supersaturation provides the driving force. Importantly, the larger population of IS-Ca^{2+} ions in confinement than on unconfined calcite could facilitate ion attachment and growth owing to their partial dehydration and stronger interaction with the surface. Nevertheless, the supersaturation for mineral growth must be higher than in free solution because the hydration force exerts a pressure on calcite that increases its solubility (35). Higher pressures are possible at asperities, which are expected on surfaces in nature, thereby leading to a local increase in the solubility; for asperities with large curvature, the influence of the interfacial energy on the solubility product must be considered as well (35). In agreement with these expectations, experiments suggest that the supersaturation in a nanoconfined solution between halite and glass during growth is greater than in the bulk (36).

Whereas ion pair attachment to calcite can happen in our hypothetical scenario, it is expected that dissolution in regions of high normal stress and subsequent precipitation at the edges of asperities, where stress and curvature are lower, will lead to a local redistribution of the mineral and of the counter- and coions within the confined solution with time (17), and thereby to changes in the hydration force and in pressure. Despite a lack of experimental data, a recent microscopic model has predicted an enhanced growth and dissolution rate of crystalline surfaces in confinement (37), which is also consistent with our picture of a dynamic growth/dissolution process at the buried calcite–solution interface. Finally, ion transport may be a crucial limiting factor in confined systems that will be investigated in the future.

Interactions that happen at the nanoconfined mineral–solution interface are central to many natural processes occurring at or near the Earth's surface, e.g., to inorganically or organically mediated calcium carbonate precipitation and its dissolution, CO_2 geologic sequestration, and adsorption of inorganic contaminants by minerals. We show that the confined calcite interface is composed of water layers and calcium ions of different hydration states, which can be distinguished by the different size of the resolved layers and

the applied work in force measurements to squeeze them out. The proposed picture is in good agreement with our knowledge about the unconfined calcite interface, but our results indicate that nanoconfinement affects the population of calcium ions of different hydration states at the calcite–solution interface. Although we focus on calcite, these results are also of relevance for many other confined mineral interfaces and for crystallization in pores.

Materials and Methods

Saturated calcium carbonate solution was prepared by equilibrating excessive calcium carbonate powder (purity ≥99.0%, Sigma-Aldrich) in nanopure water at 25 °C; the solution was filtered with a 0.2-μm nonsterile nylon membrane (Fisherbrand) before its subsequent use. Calcium chloride (purity ≥99.0%, Sigma-Aldrich) solutions at concentrations of 0, 1, 10, and 100 mM were prepared with the abovementioned saturated calcium carbonate solution and filtered before use. The calcite surfaces were prepared by cleaving Iceland Spar calcite crystals along the $(10\bar{1}4)$ plane right before use. The calcite substrates were mounted in a fluid cell filled with 2 mL $CaCl_2$ solution and equilibrated for 24 h before the force measurements, as in ref. 32. The 5-μm-radius silica microspheres (Microspheres-Nanospheres) were glued to the tipless AFM cantilevers (CSC37/tipless/Al, Mikromasch) with the aid of a micromanipulator (Sutter Instrument). The spring constant ranged between 0.2 and 0.7 N/m. The tips were soaked and cleaned with nanopure water followed by ethanol and then UV-ozone treated for 60 min immediately before the experiments. The cantilevers were equilibrated in $CaCl_2$/$CaCO_3$ solutions for 3 h before force measurements. To prepare the silica–silica systems, the silica colloids were glued on silicon wafers (p-type Boron <111> 500 μm, WRS). The substrates were sonicated in toluene, isopropanol, and ethanol (Sigma-Aldrich), three times for 15 min in each solvent. The dry silicon wafers with the colloidal spheres were UV-ozone treated for 60 min and then equilibrated for 3 h in $CaCl_2$/$CaCO_3$ solutions.

An AFM (JPK) located in an acoustic chamber was used throughout this work. Before and after each experiment, the silica colloidal probes were invert-imaged in contact mode with a calibration grid (Ted Pella) in the corresponding aqueous solutions. Direct force measurements were collected in contact mode at a speed of 20 nm/s and a maximum applied force of 4 nN. Based on the Hertz model, the contact area between calcite and silica has a radius of 7.4 nm at an applied load of 4 nN, using 0.33 and 0.17 as the Poisson ratio for calcite and silica, respectively, and 70 GPa as the Young's modulus for both. Force measurements were performed on larger atomically flat regions of the calcite crystal on the $(10\bar{1}4)$ plane. The force–separation curves were collected with a 3-min interval between each measurement to ensure the reformation of the EDL; a minimum of 100 curves was measured to collect a sufficient number of force–distance curves for statistical analysis. The fluid cell was covered with a membrane to minimize evaporation. The experiments were all conducted at 25 °C .

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